DSA06 DATA HANDLING AND VISUALIZATION

R.No: 22

Reg. No: 192425257

Name: MOTHUKURI MARUTHI

Part 1 — R (20 marks)

R1. Multiple time-series chart and growth slopes

The script draws one line per branch and labels each branch directly at the right edge, without a legend. It fits lm(Sales ~ Month_no) separately to each branch and prints the slope in Rs lakh per month. The fastest-growing branch is the one with the largest slope. Record the three values and branch name from the R console / R_Results.csv after running the script.

R2. T Nagar STL decomposition

T Nagar is converted to a monthly time series using ts(..., frequency = 12, start = c(2023, 1)) and decomposed with stl(x, s.window = "periodic"). The script plots trend, seasonal and remainder components, then reports the seasonal peak month, trough month and Fs = max(0, 1 - var(remainder)/var(seasonal + remainder)). Use the values printed by the script; they are computed from this seeded dataset.

R3. Total sales and trailing moving averages

The three branches are added month by month. The total and trailing 3-month and 12-month moving averages are plotted. Because sides = 1 is used, each moving average uses the current month and only earlier months. The 12-month average lags more because it smooths over a longer window and responds more slowly to recent changes. The script prints both moving averages at month 24 (December 2024).

R4. Detrended Tambaram ACF

The script fits a linear trend to Tambaram, takes the residuals, and plots the ACF through lag 14. The reference bound is 1.96/sqrt(36) = 0.3267. If the absolute lag-12 autocorrelation is greater than 0.3267, it is outside the approximate 95% bound and indicates a remaining yearly-cycle pattern. The exact lag-12 value and interpretation are printed by the script.

R5. Holt-Winters forecast

The first 30 observations (January 2023–June 2025) are used to fit HoltWinters. The model forecasts the next six months (July–December 2025) with prediction.interval = TRUE. The script plots history, forecast, interval limits and actual test values; it also prints MAPE and the 95% interval widths at horizons 1 and 6. MAPE is mean(abs((actual - forecast)/actual))*100.

Part 2 — Tableau Public (15 marks)

T1. Sales by month and branch

Open the generated ts_data.csv in Tableau Public. Drag Month to Columns. On the date pill, use the small dropdown arrow and select MONTH from the lower, continuous/discrete date options as demonstrated in class; do not click the plus sign. Drag Sales to Rows and Branch to Color. Hover over the December 2025 point for each line and record the branch and exact Sales value.

T2. Linear trend lines and R-squared

Open the Analytics pane, drag Trend Line onto the view and choose Linear. Ensure the trend lines are fitted per branch. Right-click a trend line and choose Describe Trend Model. Record the R-squared for each branch, then identify the highest and lowest. These values must be read from your own Tableau model; do not substitute invented values.

T3. Total sales and moving average

Create a new sheet. Put Month on Columns and SUM(Sales) on Rows; remove Branch from the view. Right-click SUM(Sales) > Quick Table Calculation > Moving Average. Configure/verify the window so it is a trailing 3-month or 12-month window only if your class instructions specify one; the prompt does not specify the window length for Tableau, so confirm the intended default with your instructor. Hover over December 2024 and record the value and the calculation window shown. Compare it with the R3 value only when both use the same window and alignment.

T4. Percent difference

On the same total-sales sheet, change the table calculation to Quick Table Calculation > Percent Difference. Hover over December 2025 and record the displayed percentage. Confirm the calculation is computed along Month and note whether Tableau shows a fraction or percentage-formatted result.

T5. Two real differences between R and Tableau

Use two differences you actually experienced. Suitable examples to verify during your work: (1) R required writing explicit code for the seeded data, STL decomposition, ACF, forecast and numerical metrics, while Tableau used drag-and-drop shelves and menus for the chart and trend lines. (2) R returned reproducible numerical outputs and saved plots/CSV from a script, while Tableau made it easier to inspect values interactively by hovering and to explore trend models visually. Rewrite these to reflect what happened in your session.

Numerical checks and answers to record

The values below are generated by the accompanying R script. Because the execution environment used to prepare this template does not have R or Tableau installed, the R-specific STL/Holt-Winters outputs and Tableau-only model values are intentionally not fabricated. Run the script and copy the exact printed values into the spaces below.